



Introduction to Machine Learning

What is Machine Learning?

“Learning is any process by which a system improves performance from experience.”

- Herbert Simon

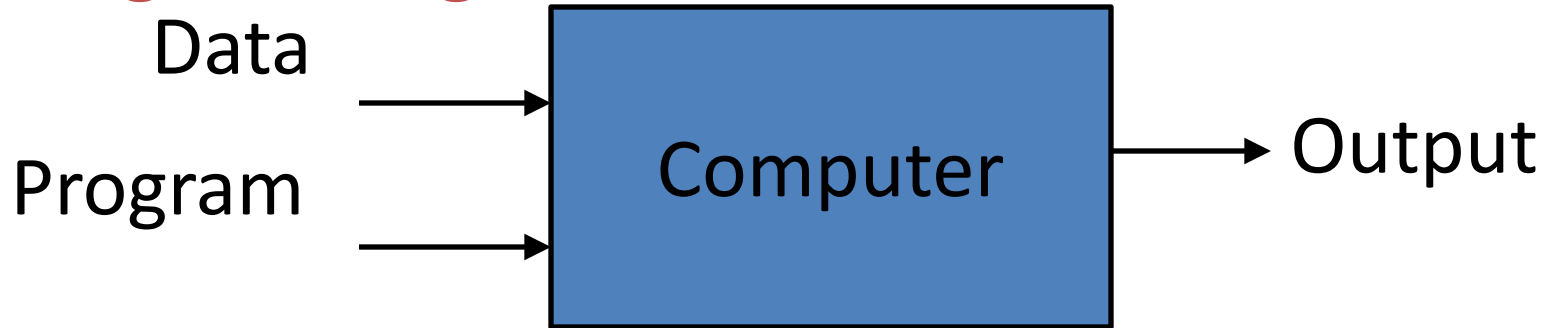
Definition by Tom Mitchell (1998):

Machine Learning is the study of algorithms that

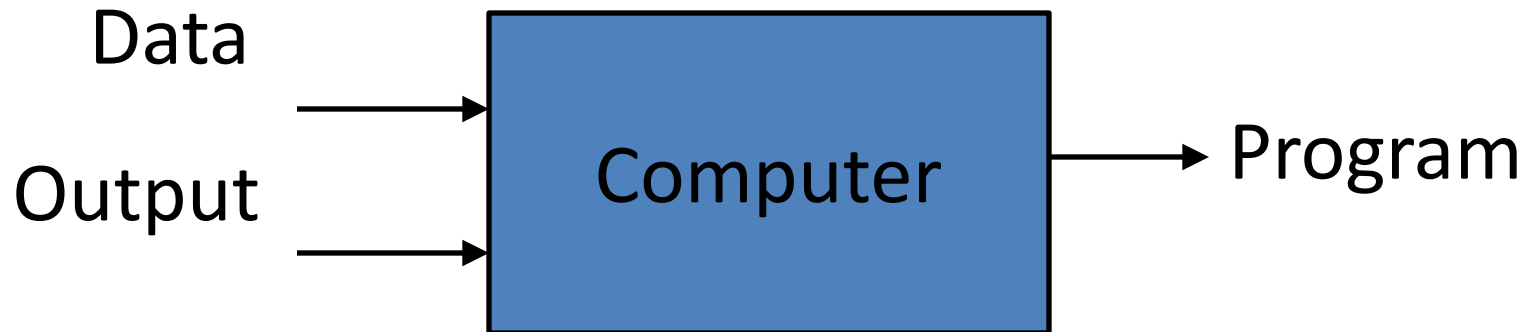
- improve their performance P
- at some task T
- with experience E .

A well-defined learning task is given by $\langle P, T, E \rangle$.

Traditional Programming



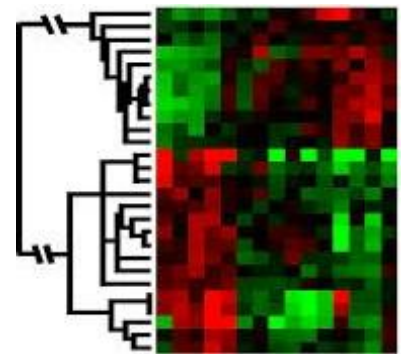
Machine Learning



When Do We Use Machine Learning?

ML is used when:

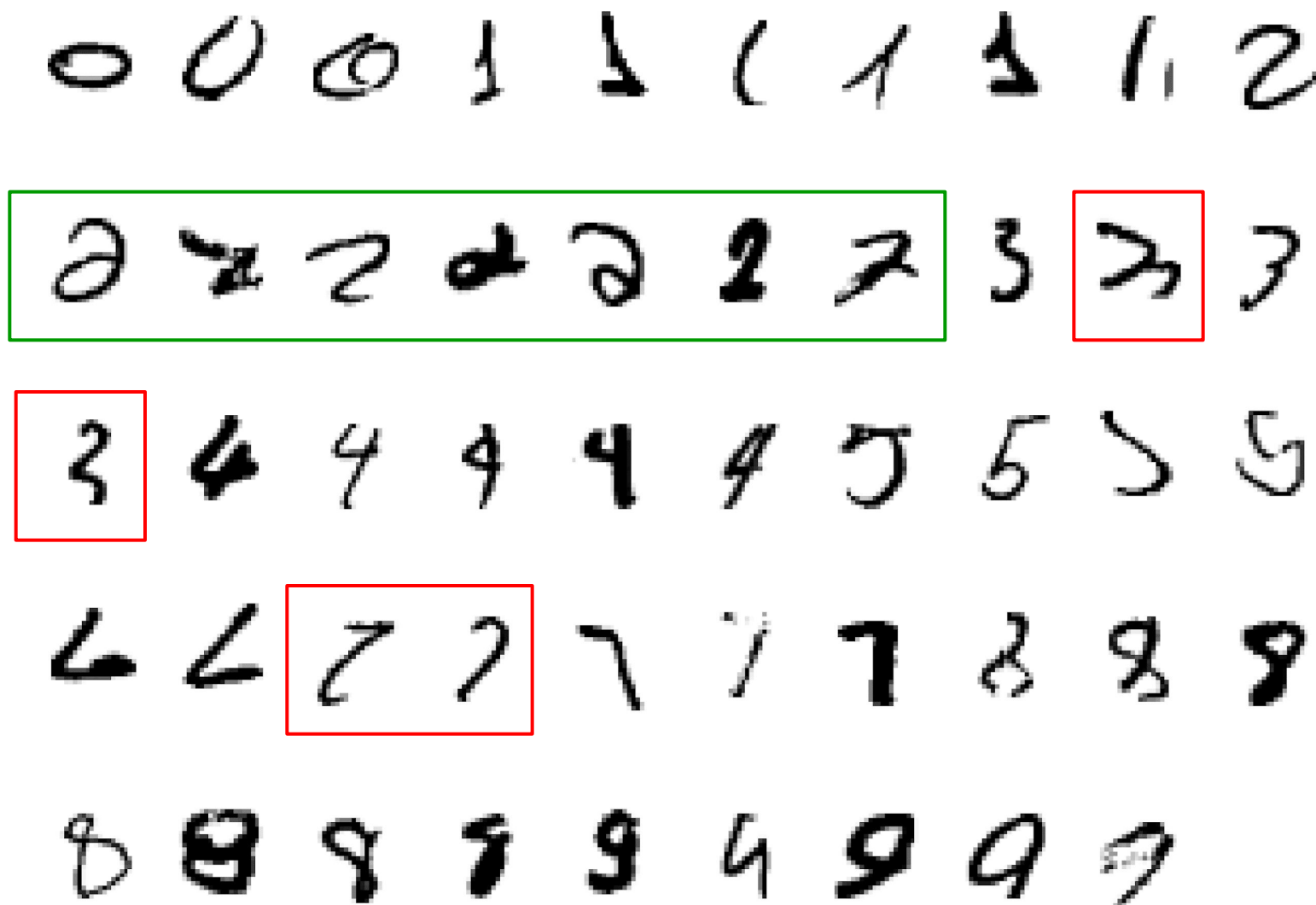
- Human expertise does not exist (navigating on Mars)
- Humans can't explain their expertise (speech recognition)
- Models must be customized (personalized medicine)
- Models are based on huge amounts of data (genomics)



Learning isn't always useful:

- There is no need to “learn” to calculate payroll

A classic example of a task that requires machine learning:
It is very hard to say what makes a 2



Some more examples of tasks that are best solved by using a learning algorithm

- Recognizing patterns:
 - Facial identities or facial expressions
 - Handwritten or spoken words
 - Medical images
- Generating patterns:
 - Generating images or motion sequences
- Recognizing anomalies:
 - Unusual credit card transactions
 - Unusual patterns of sensor readings in a nuclear power plant
- Prediction:
 - Future stock prices or currency exchange rates

Sample Applications

- Web search
- Computational biology
- Finance
- E-commerce
- Space exploration
- Robotics
- Information extraction
- Social networks
- Debugging software
- [Your favorite area]

Defining the Learning Task

Improve on task T, with respect to performance metric P, based on experience E

T: Playing checkers

P: Percentage of games won against an arbitrary opponent

E: Playing practice games against itself

T: Recognizing hand-written words

P: Percentage of words correctly classified

E: Database of human-labeled images of handwritten words

T: Driving on four-lane highways using vision sensors

P: Average distance traveled before a human-judged error

E: A sequence of images and steering commands recorded while observing a human driver.

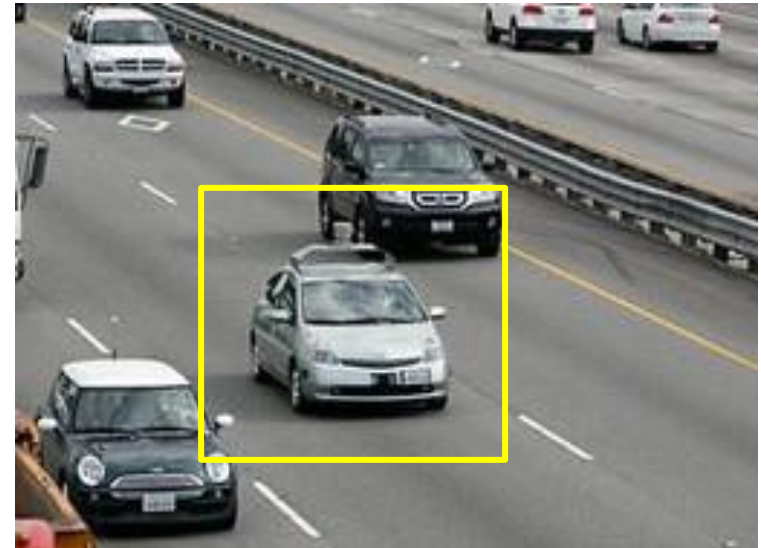
T: Categorize email messages as spam or legitimate.

P: Percentage of email messages correctly classified.

E: Database of emails, some with human-given labels

State of the Art Applications of Machine Learning

Autonomous Cars

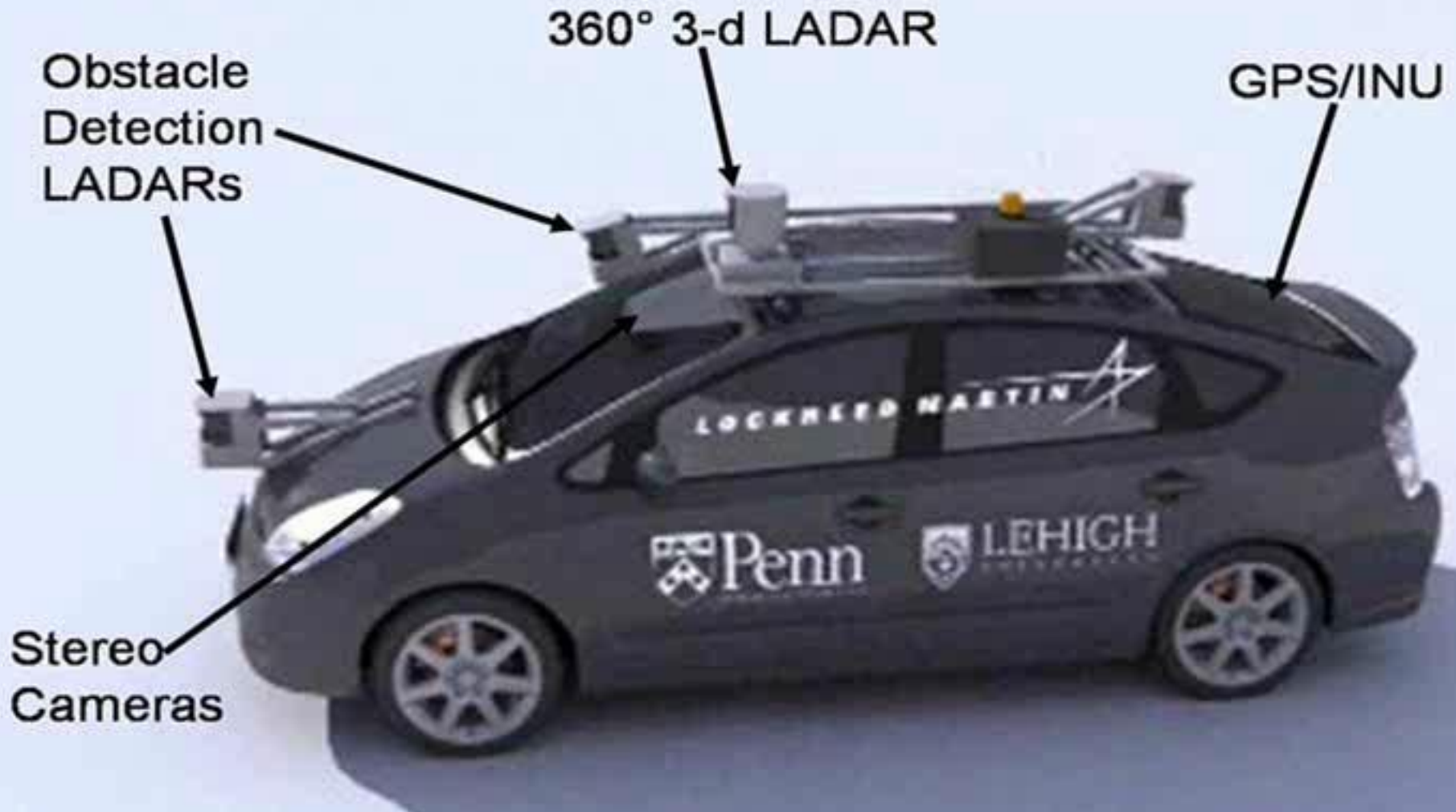


- Nevada made it legal for autonomous cars to drive on roads in June 2011
- As of 2013, four states (Nevada, Florida, California, and Michigan) have legalized autonomous cars

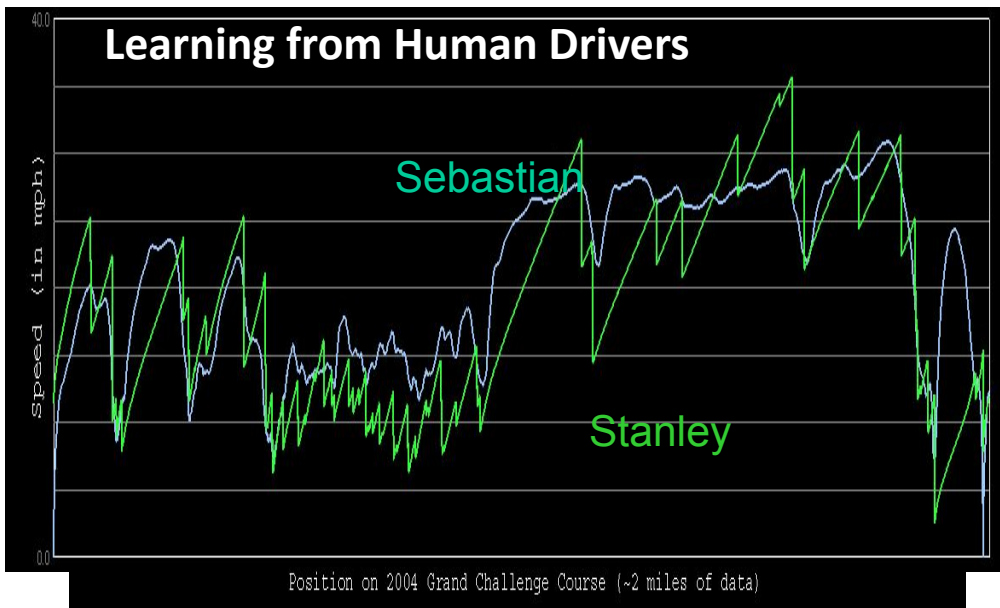
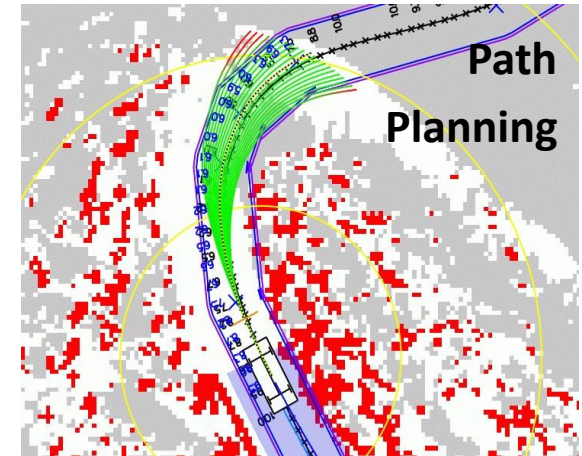
Penn's Autonomous Car □
(Ben Franklin Racing



Autonomous Car Sensors



Autonomous Car Technology

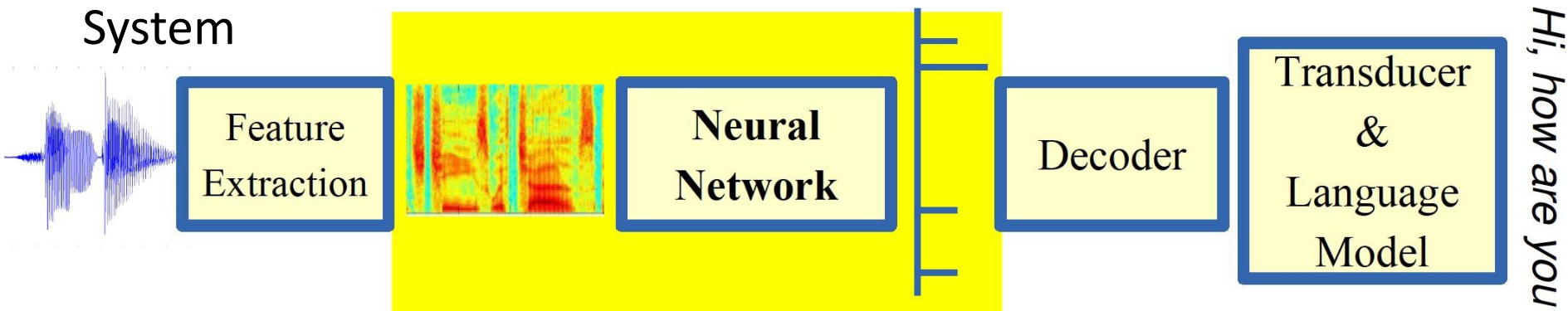


Images and movies taken from Sebastian Thrun's multimedia

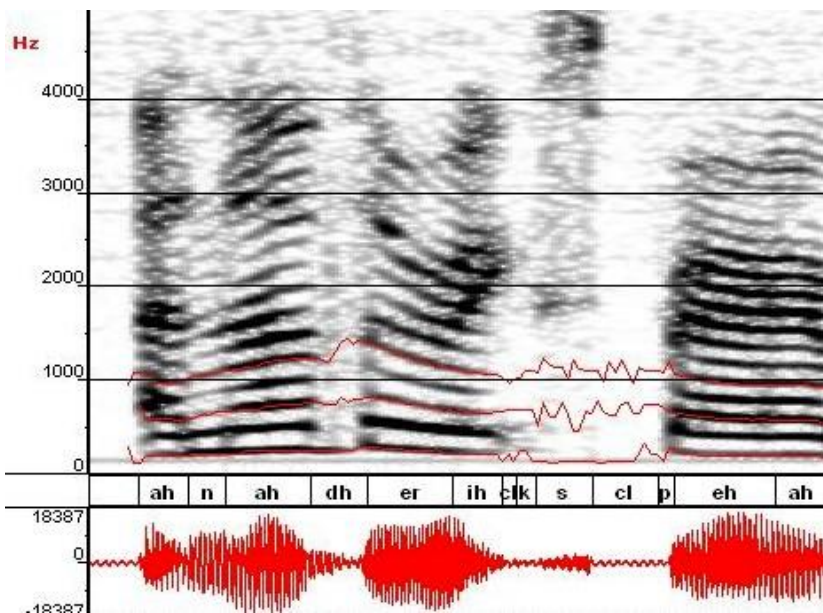
website.

Machine Learning in Automatic Speech Recognition

A Typical Speech Recognition System



ML used to predict of phone states from the sound spectrogram



Deep learning has state-of-the-art results

# Hidden Layers	1	2	4	8	10	12
Word Error Rate %	16.0	12.8	11.4	10.9	11.0	11.1

Baseline GMM performance = 15.4%

[Zeiler et al. "On rectified linear units for speech recognition" ICASSP 2013]

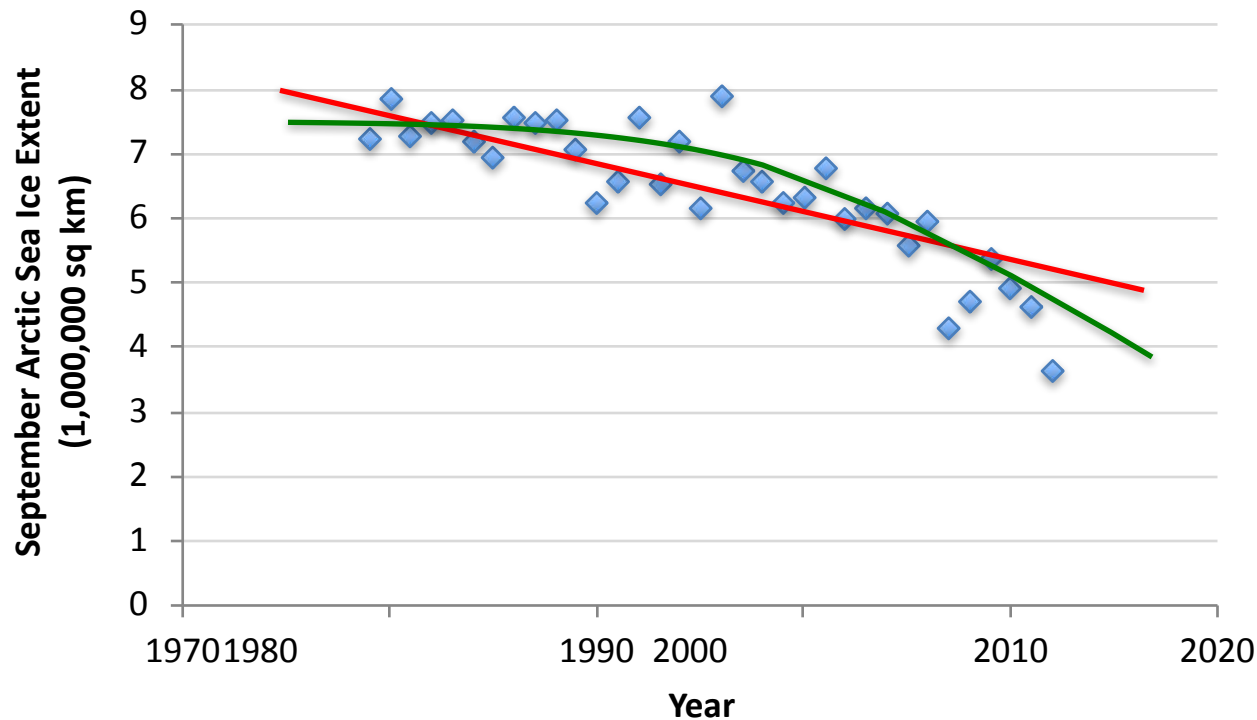
Types of Learning

Types of Learning

- **Supervised (inductive) learning**
 - Given: training data + desired outputs (labels)
- **Unsupervised learning**
 - Given: training data (without desired outputs)
- **Semi-supervised learning**
 - Given: training data + a few desired outputs
- **Reinforcement learning**
 - Rewards from sequence of actions

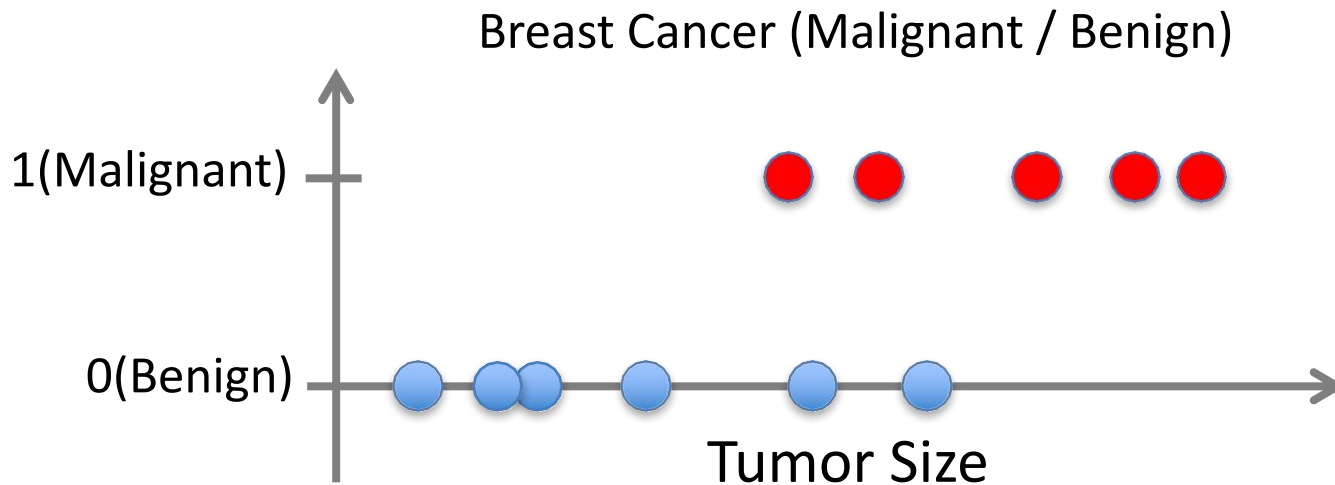
Supervised Learning: Regression

- Given $(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$
- Learn a function $f(x)$ to predict y given x
 - y is real-valued == regression



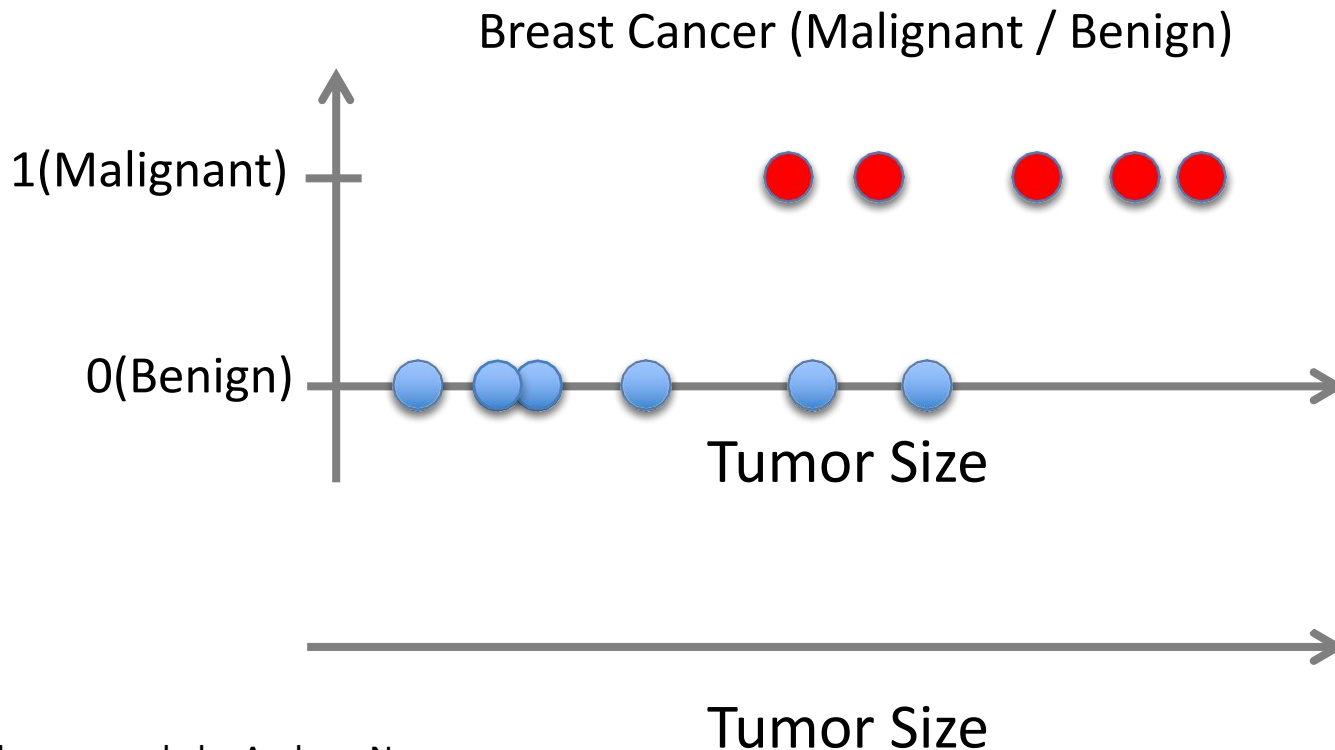
Supervised Learning: Classification

- Given $(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$
- Learn a function $f(x)$ to predict y given x
 - y is categorical == classification



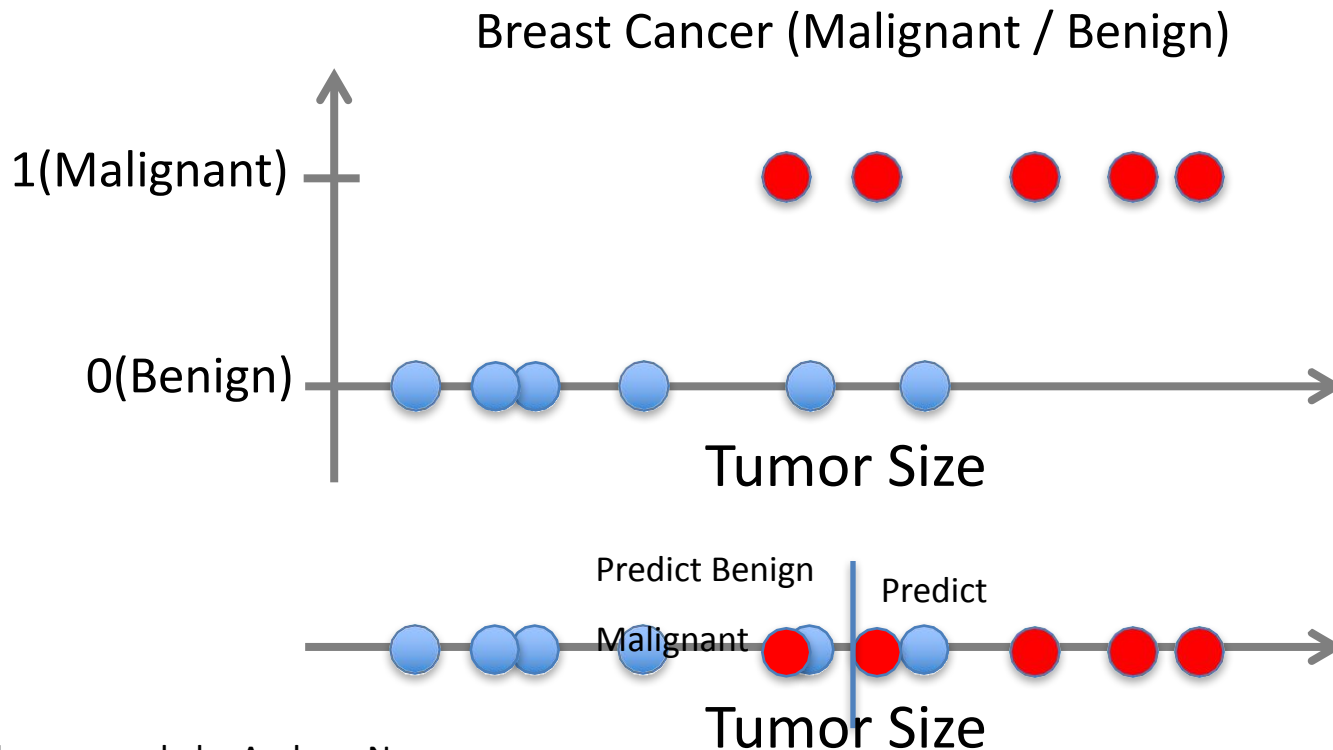
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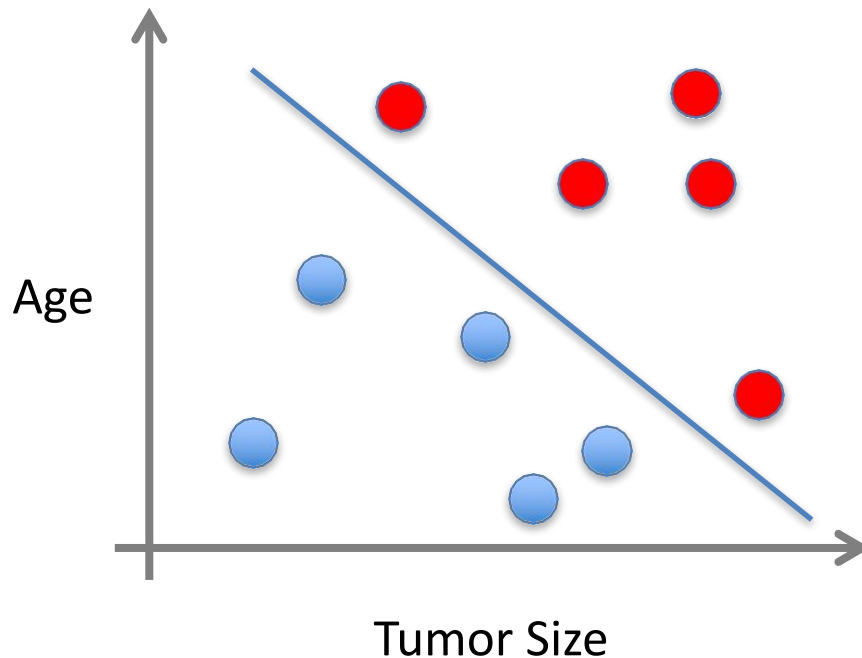
Supervised Learning: Classification

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Supervised Learning

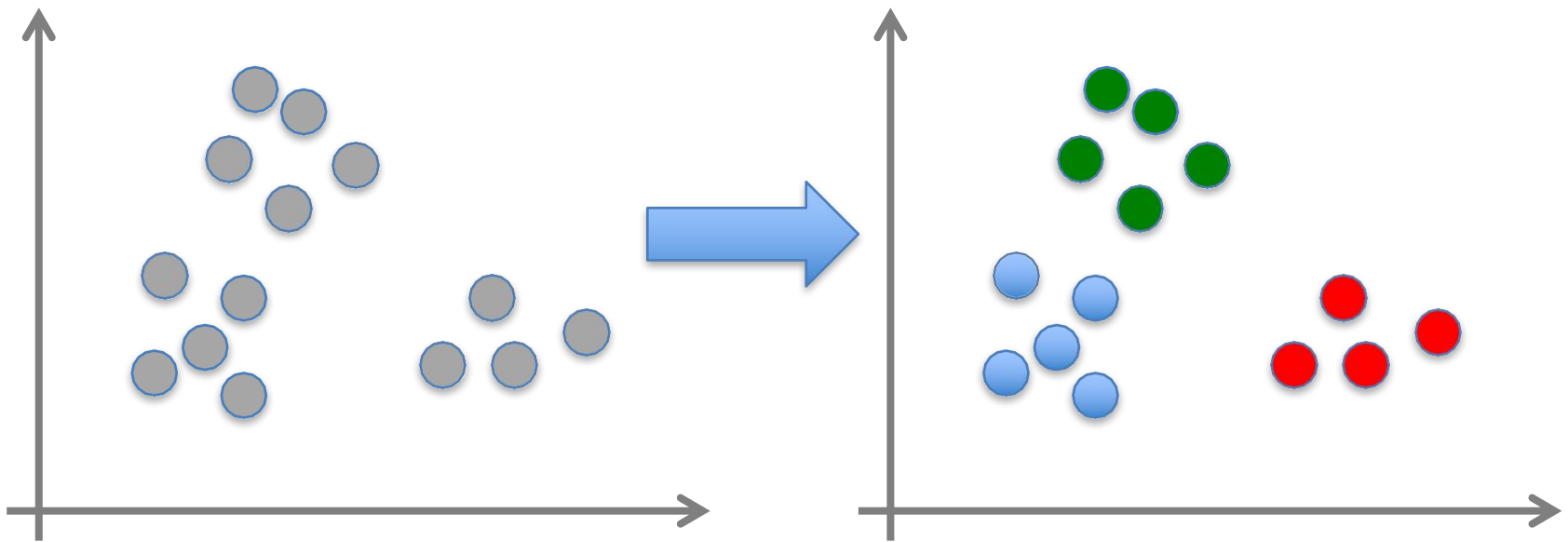
- x can be multi-dimensional
 - Each dimension corresponds to an attribute



- Clump Thickness
- Uniformity of Cell Size
- Uniformity of Cell Shape
- ...

Unsupervised Learning

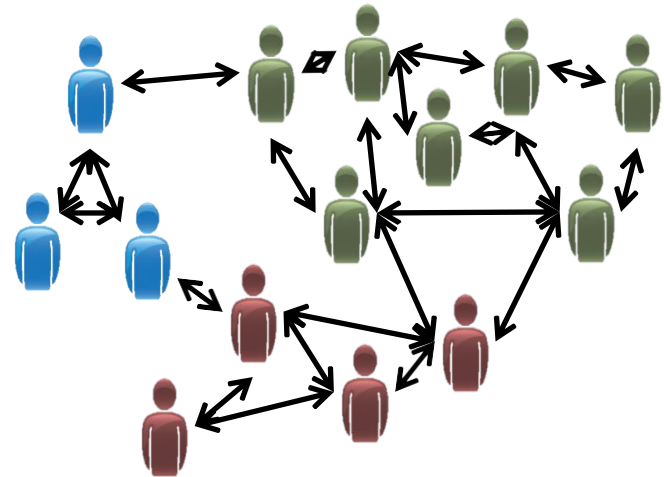
- Given $x_1, x_2, \dots, x_n$ (without labels)
- Output hidden structure behind the x 's
 - E.g., clustering



Unsupervised Learning



Organize computing clusters



Social network analysis



Market segmentation

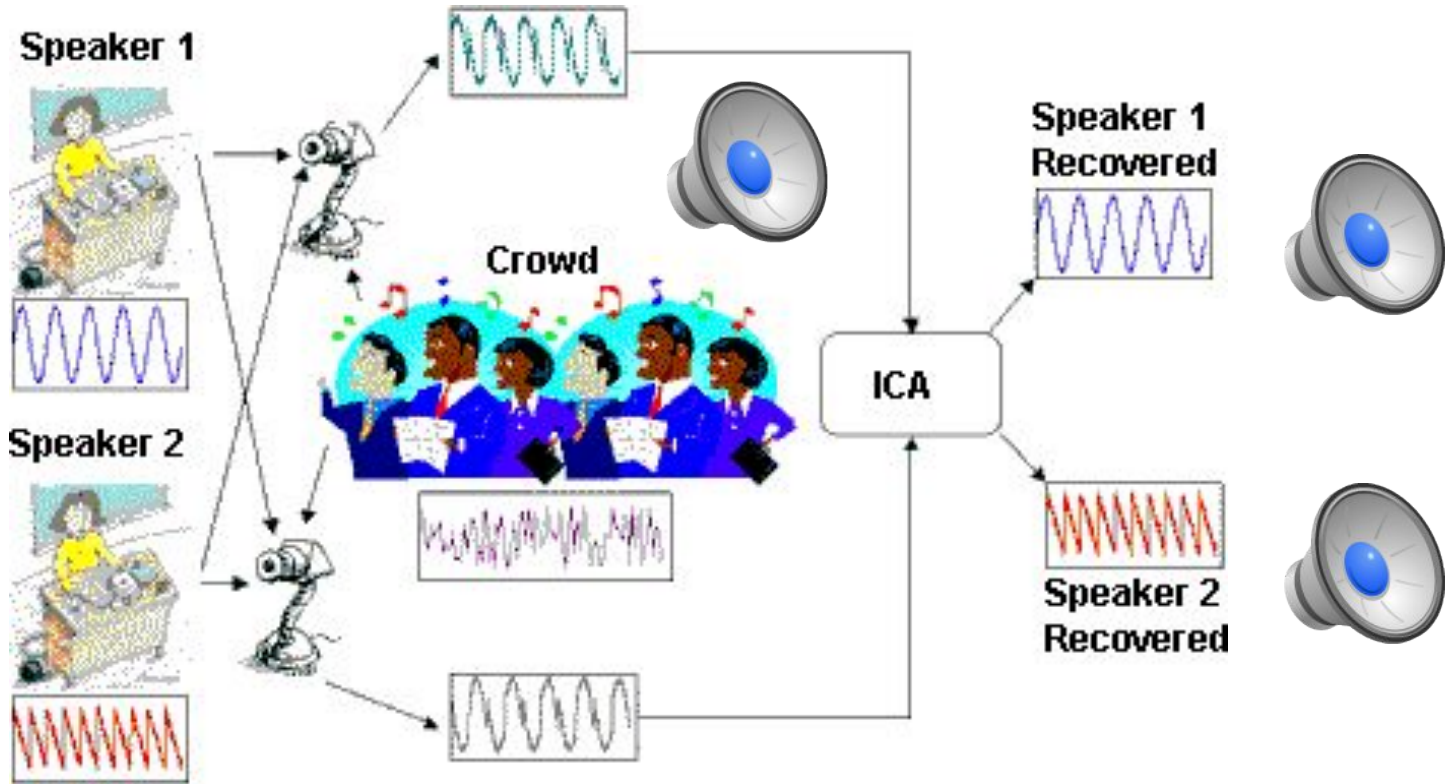


Image credit: NASA/JPL-Caltech/E. Churchwell (Univ. of Wisconsin, Madison)

Astronomical data analysis

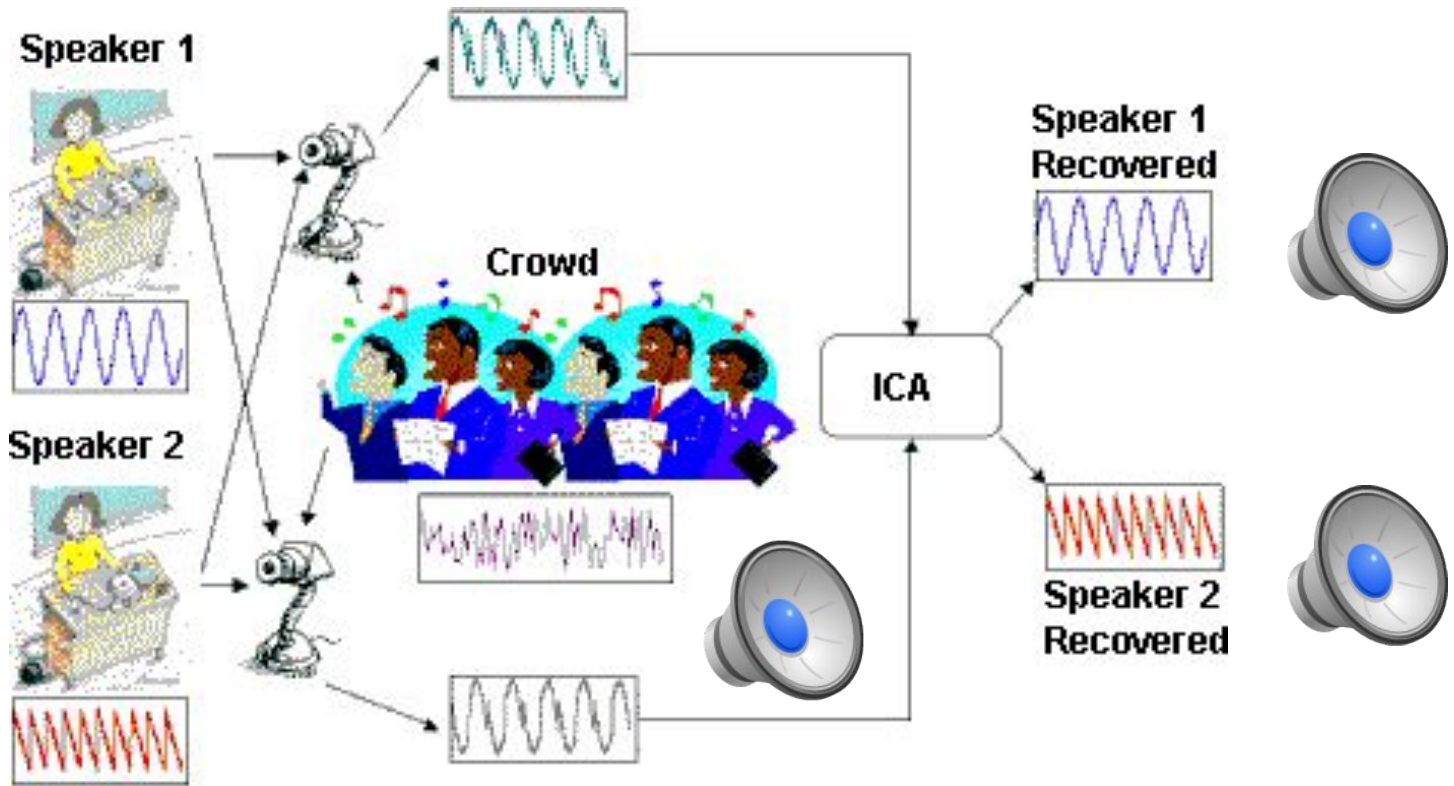
Unsupervised Learning

- Independent component analysis – separate a combined signal into its original sources



Unsupervised Learning

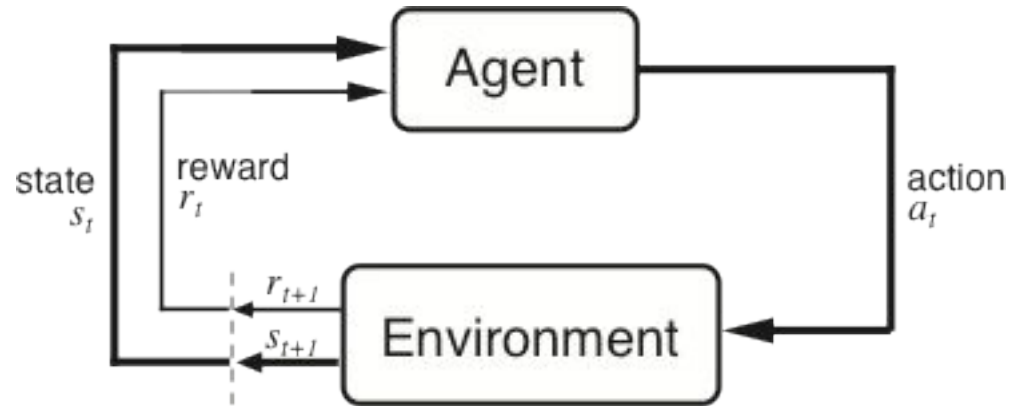
- Independent component analysis – separate a combined signal into its original sources



Reinforcement Learning

- Given a sequence of states and actions with (delayed) rewards, output a policy
 - Policy is a mapping from states $\rightarrow$ actions that tells you what to do in a given state
- Examples:
 - Credit assignment problem
 - Game playing
 - Robot in a maze
 - Balance a pole on your hand

The Agent-Environment Interface



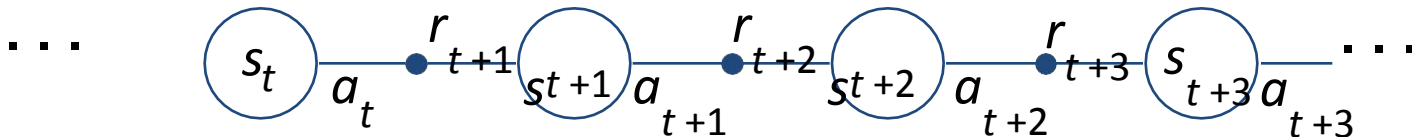
Agent and environment interact at discrete time steps : $t = 0, 1, 2, K$
 Agent observes state at step t : $s_t \in S$

produces action at step t : $a_t \in A(s_t)$

gets resulting reward : $r_{t+1} \in \mathcal{R}$

and resulting next state s_{t+1}

:

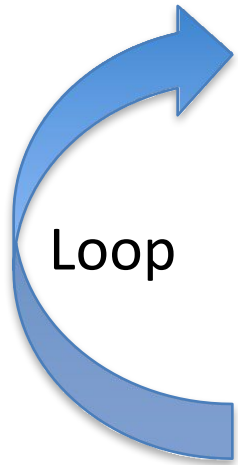


Reinforcement Learning



<https://www.youtube.com/watch?v=4cgWya-wjgY>

ML in Practice



- Understand domain, prior knowledge, and goals
- Data integration, selection, cleaning, pre-processing, etc.
- Learn models
- Interpret results
- Consolidate and deploy discovered knowledge